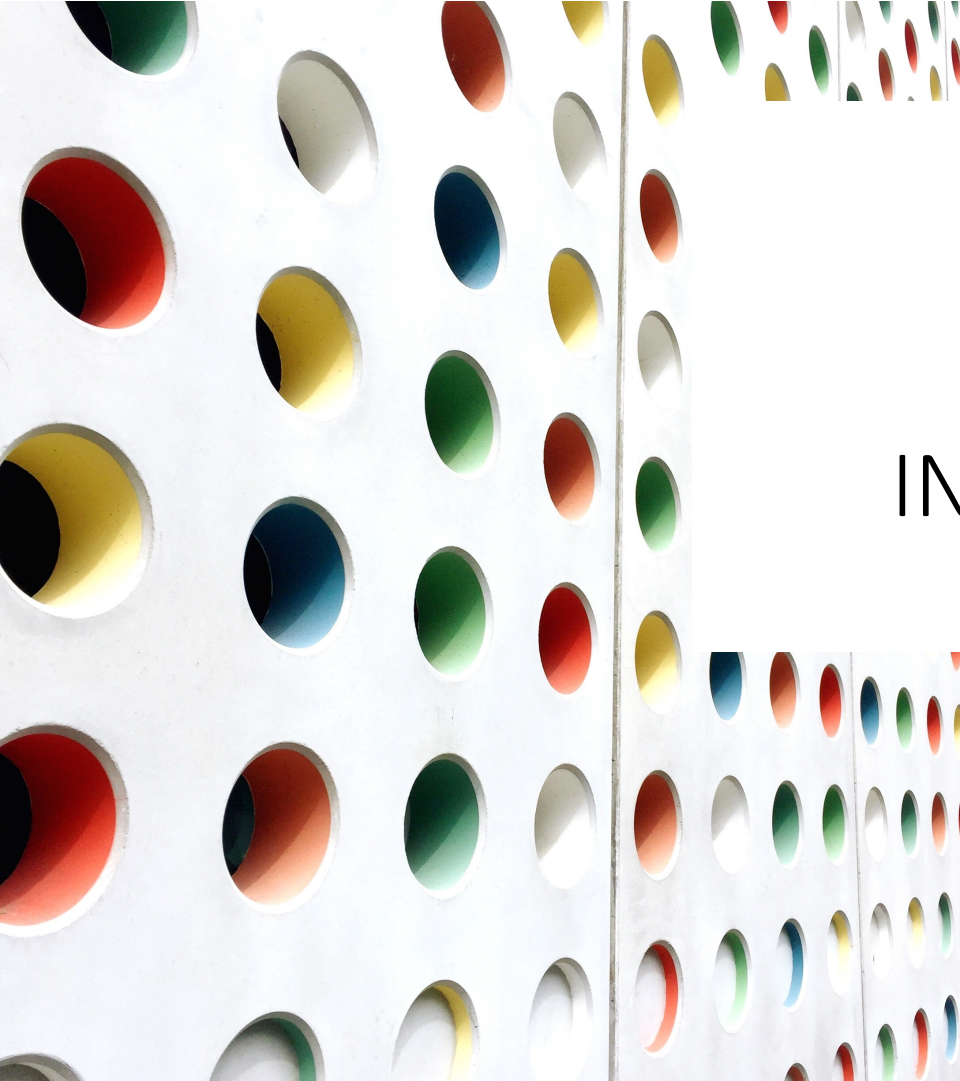




ISP550 INFORMATION SYSTEMS ENGINEERING

Topic 4: **Project Planning and
Initiation**

OKTOBER 2025 – FEBRUARY 2026

Learning Objectives

- Describe the **project identification** and **selection process**
- Describe **the steps** involved in the project initiation and planning process
- List and describe various methods for **assessing project feasibility**
- Describe the **classes of Internet electronic commerce applications**: business-to-consumer, business-to-employee, and business-to-business

Figure 3.6 – The Project Life Cycle (PLC) and the Systems Development Life Cycle (SDLC)

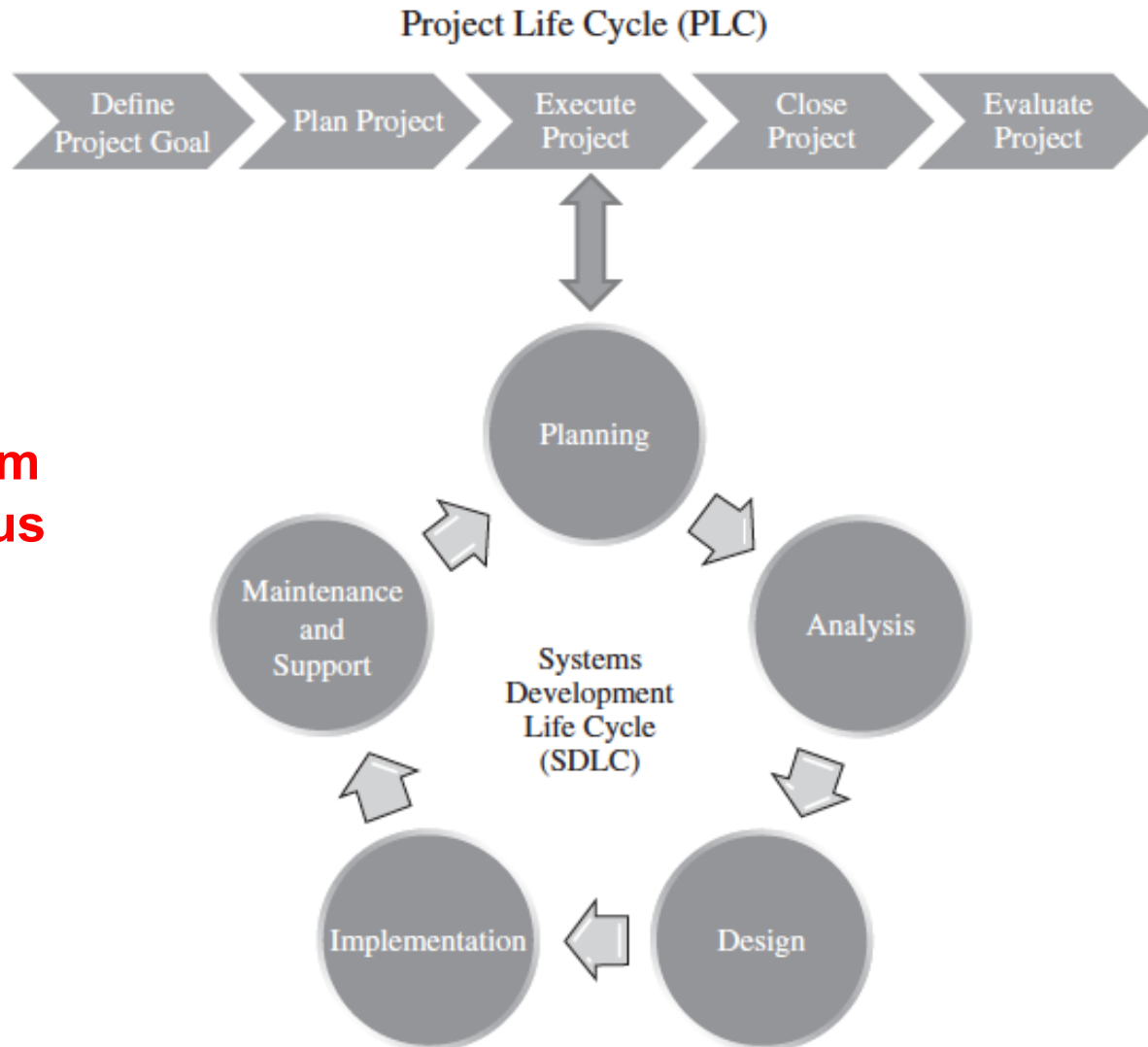
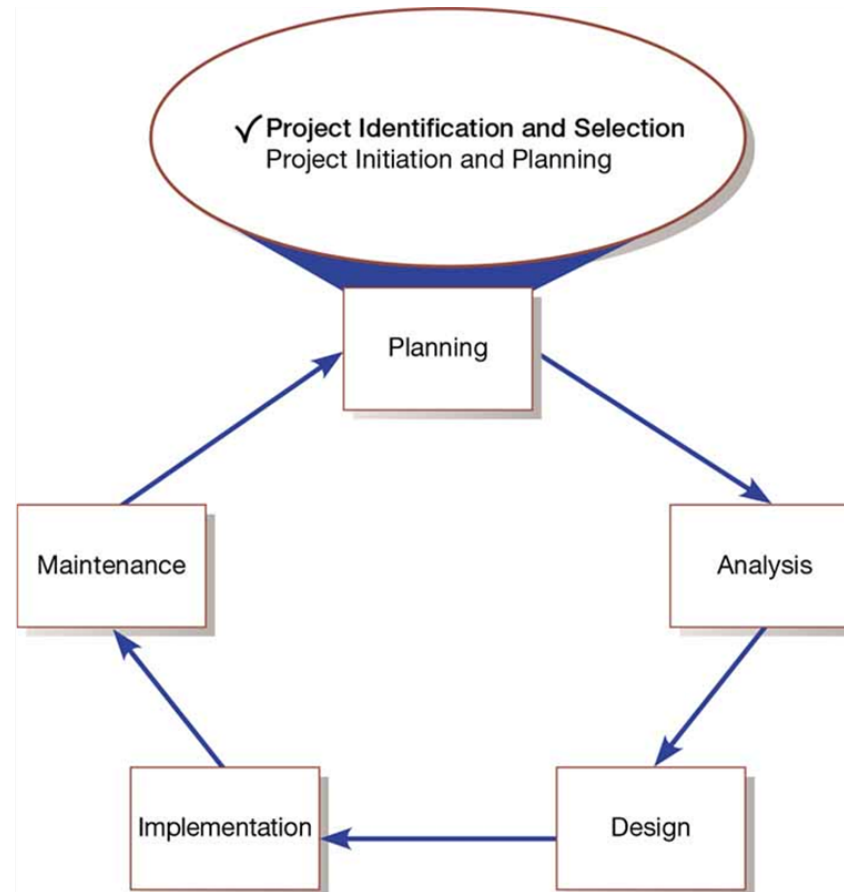


Figure From
the previous
topic

Introduction

- Obtaining **integrated, enterprise-wide computing** presents significant challenges for both corporate and information system management
- The **acquisition, development, and maintenance** of information systems consume resources for most organizations
- This leads to the need to have a formal process for identifying and selecting projects

Figure 4-1: Systems Development Life Cycle with Project Identification and Selection Highlighted



1. Identifying and Selecting Systems Development Projects

- Identifying and selecting systems development projects is part of the **first phase (Planning)** of the SDLC
- Many firms follow a very **formal process** for selecting projects
- **Requests come from** a variety of sources:
 - By managers needing to **replace aging existing systems**
 - By managers wanting to **make a system more efficient**
 - By managers **needing a new system**

Identifying and Selecting Projects (1 of 3)

- Project identification and selection consists of three primary activities:
 1. Identifying potential development projects
 - By key member of top management
 - By steering committee (**top-down source**)
 - By user departments (**bottom-up source**)
 - By development group or senior IS manager

Identifying and Selecting Projects (2 of 3)

2. Classifying and ranking IS development projects

- Using value chain analysis or other evaluation criteria
 - **Value chain analysis** – used in analyzing an organization's activities to determine where value is added to products and/or services and the costs incurred for doing so; it usually also includes a comparison with the activities, added value, and costs of other organizations for the purpose of making improvements in the organization's operations and performance

Identifying and Selecting Projects (3 of 3)

3. Selecting IS development projects

Selection Method	Characteristics
Top Management	Greater strategic focus Largest project size Longest project duration Enterprise-wise consideration
Steering Committee	Cross-functional focus Greater organizational change Formal cost-benefit analysis Larger and riskier projects
Functional Area	Narrow, nonstrategic focus Faster development Fewer users, management layers, and business functions involved
Development Group	Integration with existing systems focus Fewer development delays Less concern with cost-benefit analysis

Table 4-1: Characteristics of Alternative Methods for Making Information Systems Identification and Selection Decisions

Table 4-2: Possible Evaluation Criteria When Classifying and Ranking Projects

Evaluation Criteria	Description
Value Chain Analysis	Extent to which activities add value and costs when developing products and/or services
Strategic Alignment	Extent to which the project is viewed as helping the organization achieve its strategic objectives and long-term goals
Potential Benefits	Extent to which the project is viewed as improving profits, customer service, and so forth, and the duration of these benefits
Resource Availability	Amount and type of resources the project requires and their availability
Project Size/Duration	Number of individuals and the length of time needed to complete the project
Technical Difficulty/Risks	Level of technical difficulty to successfully complete the project within given time and resource constraints

Figure 4-1: Organizations Can Be Thought of as a **Value Chain**, Transforming Raw Materials into Products for Customers



(**Sources:** Left to right: Bram van Broekhoven/Shutterstock; Alexey Fursov/Shutterstock; Hachob/Shutterstock; Syda Productions/Shutterstock)

Figure 4-2: Project Selection Decisions Must Consider Numerous Factors and Can Have Numerous Outcomes

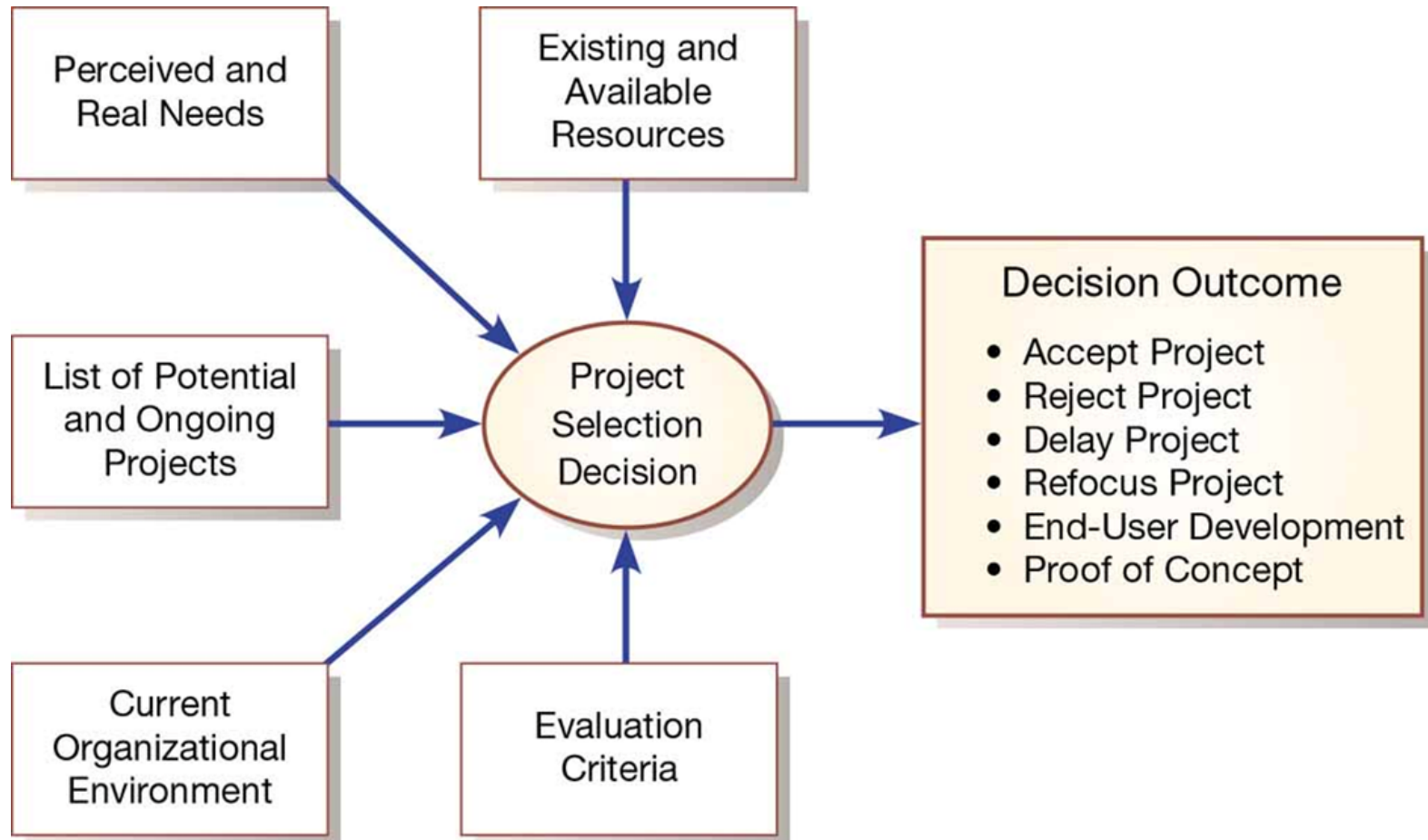


Table 4-3: Elements of Project Initiation

Elements of Project Initiation
• Establishing the project initiation team
• Establishing a relationship with the customer
• Establishing the project initiation plan
• Establishing management procedures
• Establishing the project management environment and project workbook
• Developing the project charter

#Refer Topic 3, slide 12

Table 4-4: Elements of Project Planning

Elements of Project Planning
• Describing the project scope, alternatives, and feasibility
• Dividing the project into manageable tasks
• Estimating resources and creating a resource plan
• Developing a preliminary schedule
• Developing a communication plan
• Determining project standards and procedures
• Identifying and assessing risk
• Creating a preliminary budget
• Developing the project scope statement
• Setting a baseline project plan

Deliverables and Outcomes (1 of 2)

- **Business case** – justification for an information system, presented in terms of the tangible and intangible economic benefits and costs and the technical and organizational feasibility of the proposed system
- **Baseline Project Plan (BPP)** – major outcome and deliverable from the project initiation and planning phase that contains the best estimate of a project's scope, benefits, costs, risks, and resource requirements
- **Project Scope Statement (PSS)** – document prepared for the customer that describes what the project will deliver and outlines generally at a high level all work required to complete the project

2. Initiation and Planning Projects

- When does project initiation and planning (PIP) end and execution begin?
- Three important questions must be considered when making this decision:
 - **How much effort** should be expended on the project initiation and planning phase?
 - **Who is responsible** for performing the project initiation and planning process?
 - **Why is** project initiation and planning such a **challenging activity**?

Assessing Project Feasibility

- **Most feasibility factors** are represented by the following categories:
 - Economic
 - Technical
 - Operational
 - Scheduling
 - Legal and contractual
 - Political

Assessing Economic Feasibility

- **Economic feasibility** – process of identifying the financial benefits and costs associated with a development project
- **Tangible benefit** – benefit from the creation of an information system that can be measured in dollars and with certainty
 - Most tangible benefits:
 - Cost reduction and avoidance
 - Error reduction
 - Increased flexibility
 - Increased speed of activity
 - Improvement of management planning and control
 - Opening new markets and increasing sales opportunities
- **Intangible benefit** – benefit derived from the creation of an information system that cannot be easily measured in dollars or with certainty (See Table 4-5)

Table 4-5: Intangible Benefits from the Development of an Information System

Intangible Benefits from the Development of an IS	Intangible Benefits from the Development of an IS
• Competitive necessity	• More confidence in decision quality
• More timely information	• Improved processing efficiency
• Improved organizational planning	• Improved asset utilization
• Increased organizational flexibility	• Improved resource control
• Promotion of organizational learning and understanding	• Increased accuracy in clerical operations
• Availability of new, better, or more information	• Improved work process that can improve employee moral or customer satisfaction
• Ability to investigate more alternatives	• Positive impacts on society
• Faster decision making	• Improved social responsibility
• Better usage of resources (“greener”)	

(Source: Based on Parker & Benson, 1988; Brynjolfsson & Yang, 1997; Keen, 2003; Cresswell, 2004)

Assessing Technical Feasibility

- **Technical feasibility** – process of assessing the development organization's ability to construct a proposed system
- Potential consequences of not accessing and managing risks can include the following:
 - Failure to attain expected benefits from the project
 - Inaccurate project cost estimates
 - Inaccurate project duration estimates
 - Failure to achieve adequate system performance levels
 - Failure to adequately integrate the new system with existing hardware, software, organizational procedures

Assessing Other Feasibility Concerns

- **Operational feasibility** – process of assessing the degree to which a proposed system solves business problems or takes advantage of business opportunities
- **Schedule feasibility** – process of assessing the degree to which the potential time frame and completion dates for all major activities within a project meet organizational deadlines and constraints for affecting change
- **Legal and contractual feasibility** – process of assessing potential legal and contractual ramifications due to the construction of a system
- **Political feasibility** – process of evaluating how key stakeholders within the organization view the proposed system

ELECTRONIC COMMERCE APPLICATIONS: IDENTIFYING AND SELECTING SYSTEMS DEVELOPMENT PROJECTS

- Identifying and selecting systems development projects for an Internet-based electronic commerce application is no different from the process followed for more traditional applications.
- Nonetheless, there are **some special considerations when developing an Internet-based application.**
- In this section, we highlight some of those issues that relate directly to the process of identifying and selecting Internet-related systems development projects.

Internet Basics

- **Internet** – large, worldwide network of networks that use a common protocol to communicate with each other
- **Internet of Things (IoT)** – broad class of physical objects that feature an Internet address and connectivity that communicate between these objects and other Internet enabled devices and systems
- **Electronic commerce (EC)** – Internet-based communication to support day-to-day government, business, and consumer activities

EC Business Models

- A broad range of **business models** include:
 - Business-to-Business (B2B)
 - Business-to-Consumer (B2C)
 - Consumer-to-Consumer (C2C)
 - Consumer-to-Business (C2B)
 - Business-to-Government (B2G)
 - Government-to-Business (G2B)
 - Government-to-Citizen (G2C)
 - Thing-to-Thing (T2T)

Figure 4-3: The Internet of Everything

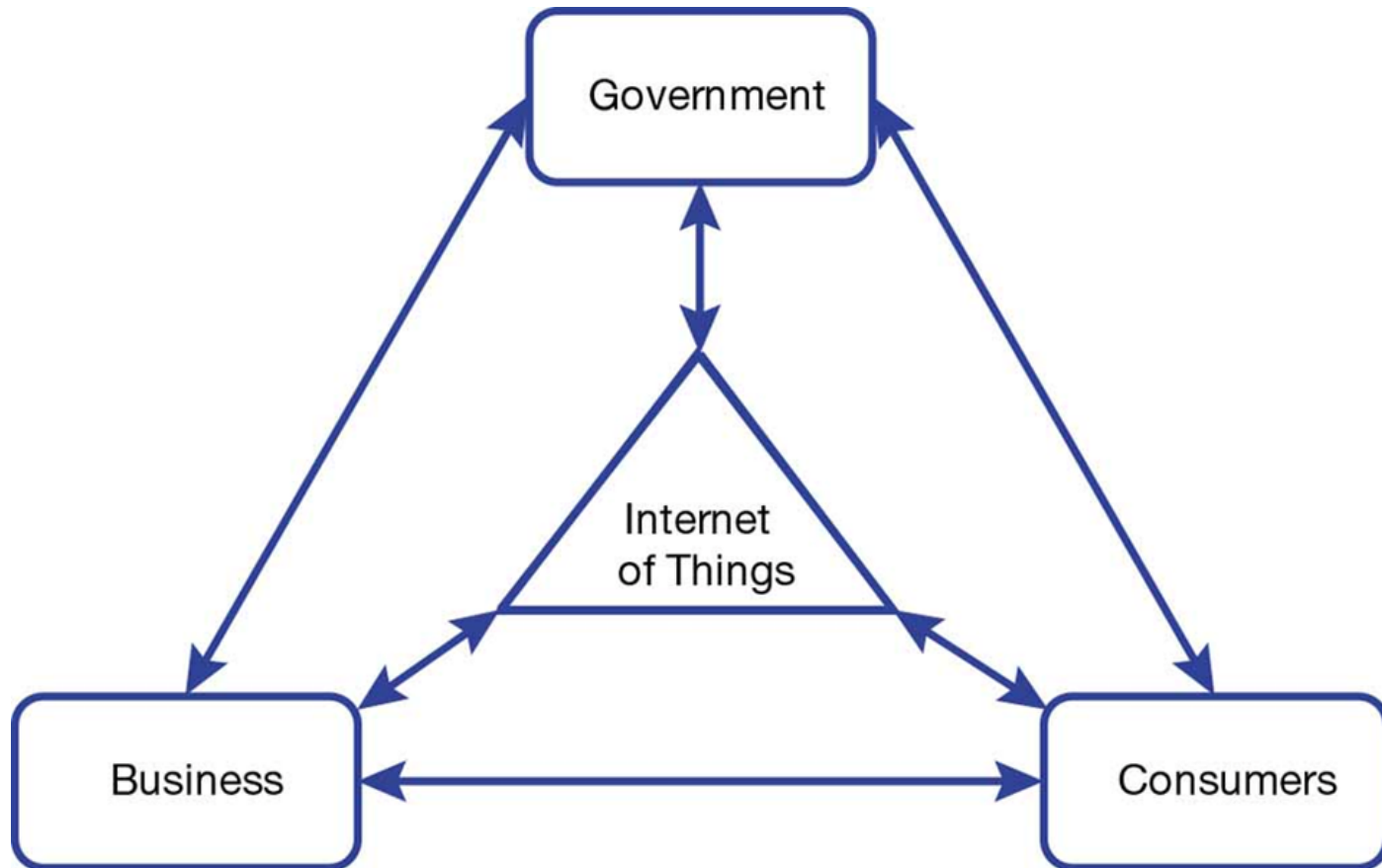
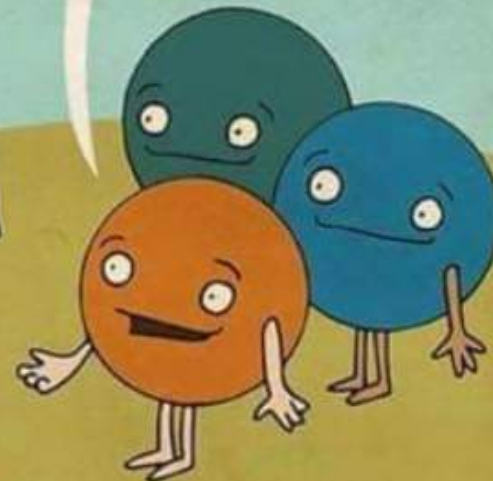
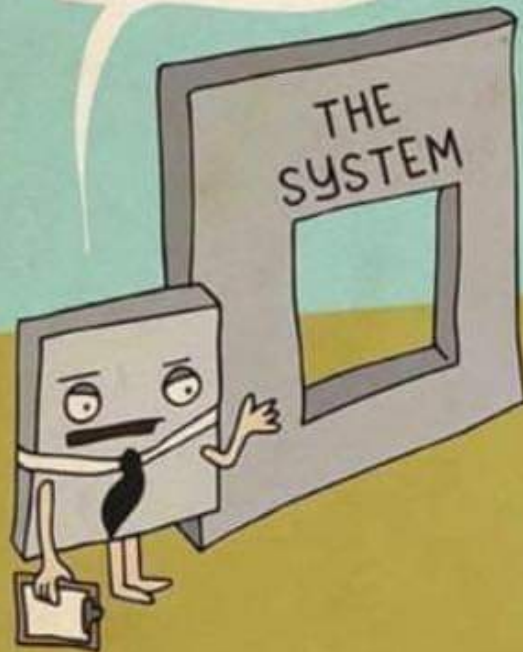


Table 4-6: Unknowns That Must Be Dealt with When Designing and Building Internet Applications

Factor	Unknowns That Must Be Dealt With
User	• Concern: Who is the user? • Example: Where is the user located? What is the user's expertise or education? What are the user's expectations?
Connection Speed	• Concern: What is the speed of the connection and what information can be effectively transported? • Example: WiFi, cellular
Access Method	• Concern: What is the connection device? • Example: Web browser, tablet, smartphone, smart watch

You're going to have
to change to fit
into the system.

Or how about you
change the system
so we can all fit?



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- In this chapter you learned to:
 - ✓ Describe the project identification and selection process
 - ✓ Describe the steps involved in the project initiation and planning process
 - ✓ List and describe various methods for assessing project feasibility
 - ✓ Describe the classes of Internet electronic commerce applications: business-to-consumer, business-to-employee, and business-to-business